

Zach Landry

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Boston, Massachusetts

Education

Northeastern University, Boston, MA

September 2024 - April 2028

Candidate for Bachelor of Science in Electrical Engineering and Physics

GPA 4.0 | Dean's List

Relevant Coursework: Fundamentals of Electronics, Embedded Design, Computing Fundamentals, Calculus III, Differential Equations & Linear Algebra, Quantum Mechanics, Electricity & Magnetism 1

Current Coursework: Electronic Design, Linear Systems, Classical Dynamics, Advanced Quantum Mechanics

Activities: IEEE, Generate Electrical Engineer, College of Engineering Tutoring, Downhillers, NUHOC

Skills

Electronics: Breadboarding, Soldering (THT/SMD), Analog Circuit Design, Embedded Hardware Design, PCB layout, DMM, Oscilloscope, Function Generator

Software: KiCad, LTspice, AutoCAD, SolidWorks, C++, Python, Arduino, MATLAB, Excel

Work Experience

Electrical Engineer | Generate | Boston, Massachusetts

September 2026 - Present

- Design a multi-axis motor and sensing system integrating steppers, BLDCs, encoders, and limit switches in collaboration with a multidisciplinary team
- Develop motor-specific hardware, including custom PCBs for power delivery, encoder interfacing, and motion-system safety

Electrical Engineering Co-op | Massa | Hingham, Massachusetts

January 2026 - June 2026

- Designed and built an ultra-low-power PIC-based embedded system on a custom PCB, integrating an external custom designed sensor module for field data acquisition
 - Developed a multi-stage analog filter chain to condition sensor signals for high-accuracy ADC sampling
 - Diagnosed and resolved complex PCB failures using oscilloscopes, signal tracing, and component-level debugging
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Personal Projects

Analog Synthesizer

May 2026 - Present

- Designed a triangle-core function generator feeding a state-variable VCF and an OTA-based VCA
- Built an ADSR envelope generator and exponential converter for control-voltage shaping
- Designed an analog AGC stage for consistent output amplitude across frequencies

Hexapod

May 2025 - July 2025

- Designed and built a 3D-printed hexapod robot
- Wired 18 servo motors to an ESP-32 microcontroller with Adafruit drivers
- Coded a gait algorithm for multi-directional movement in the Arduino IDE (written in C++)

Differential Equation Visualizer

February 2025 - April 2025

- Built an interactive frontend for visualizing ODE's
- Implemented a backend solver which supports arbitrary equation inputs
- Included slope fields for first-order equations and special step and impulse functions